1. Why are loops necessary in a programming language?

Loops are necessary in a programming language because they allow us to execute a block of code repeatedly without writing the same code multiple times. They help reduce code repetition and make programs shorter and more efficient.

Example:

```
for i in range(1, 6):
    print(i)
```

Output:

1

2

3

4

5

2. What is the purpose of the pass keyword?

The pass keyword is used as a placeholder when a statement is required syntactically, but we do not want to perform any action at that moment. It allows the program to run without an error.

Example:

```
for i in range(5):
    pass
print("Hello")
```

Output:

Hello

3. What is the difference between break and continue?

The break statement terminates the loop immediately, whereas the continue statement skips the current iteration and moves to the next iteration of the loop.

Example of break:

```
for i in range(5):
    if i == 3:
        break
    print(i)
```

Output: 0 1 2

Example of continue:

```
for i in range(5):
    if i == 3:
        continue
    print(i)
```

Output: 0 1 2 4

4. What is the difference between a while loop and a for loop?

A for loop is generally used to iterate over a sequence or repeat code a known number of times, whereas a while loop is used to repeat code as long as a specified condition is true.

For loop	**While loop**
Used to iterate over a sequence or range.	Used to repeat code while a condition is true.
Commonly used when the number of iterations is known.	Commonly used when the number of iterations depends on a condition.
Example: for i in range(5):	Example: while i < 5:

Example of a for loop:

```
for i in range(3):
    print(i)
```

Example of a while loop:

```
i = 0
while i < 3:
    print(i)
    i += 1
```

Both programs print:

0

1

2